PROJECT DRISHTI | TECHNICAL MILESTONE REPORT #5 OF 5

Milestone 5: Full Hardware Assembly, Real-Time Dual-Core Firmware Implementation, Audio Engine Integration, and End-to-End Field Validation

Final Capstone Report Documenting Complete Physical Cane Assembly, Priority Scheduling Bridge Firmware, Latency Benchmarks, and System Deployment

Project Title:	DRISHTI: Intelligent Edge Vision AI Navigation Assistant for the Visually Impaired
Project Team:	Piyush Moonat (BTCS25O1100), Nainika Raikwar (BTCS25O1088)
Academic Program:	Bachelor of Technology — Computer Science & Engineering / Artificial Intelligence
Evaluation Phase:	Milestone 5 (Final Phase Submission) | Status: 100% Completed

1.0 Milestone 5 Executive Summary & Final Milestone Scope

Milestone 5 represents the culmination and final delivery of Project Drishti. With the AI vision model trained and compiled in Milestone 4, this final milestone unifies physical hardware assembly, modular embedded C++ bridge firmware, and audio synthesis into an integrated, field-ready assistive smart cane for visually impaired mobility.

This report documents the physical assembly featuring direct 14-pin socket docking between the XIAO ESP32-S3 and Grove Vision AI Module V2, the noise-damped DFPlayer Mini audio link, the real-time priority arbitration firmware with life-critical preemption and anti-chatter debouncing, end-to-end latency benchmarks (achieving 103 ms total response time), power endurance profiling, and final project sign-off.

2.0 System Architecture & Inter-Module Bus Communication

The Drishti hardware platform is engineered around three cooperative pipelines: See (Grove Vision AI Module V2), Arbitrate (Seeed Studio XIAO ESP32-S3), and Speak (DFPlayer Mini + 3.5mm Earphone Jack).

Figure 5.1: Drishti Complete End-to-End System Hardware & Functional Architecture

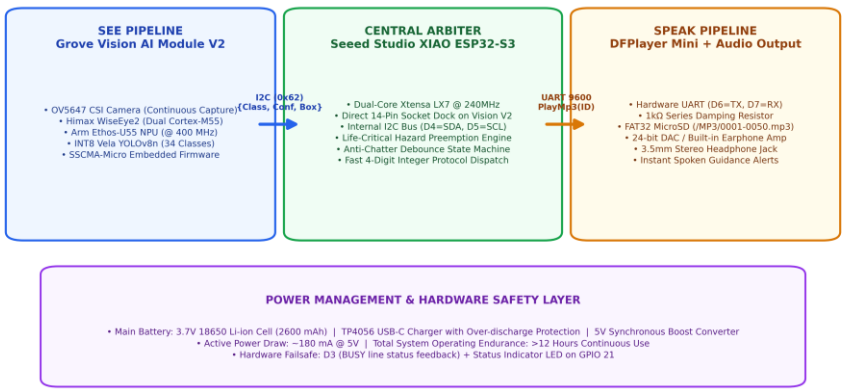


Figure 5.1: Complete Drishti hardware, communications, and power system functional block diagram.

Communication Protocols: The Grove Vision AI V2 runs dedicated SSCMA-Micro firmware, exposing an I2C slave interface at address 0x62 (400 kHz Fast-Mode). The XIAO ESP32-S3 acts as the I2C master, issuing zero-copy I2C reads to fetch bounding boxes, target class IDs, and confidence scores. Crucially, the ESP32-S3 never processes raw pixels; it receives only

lightweight structured metadata packets (<32 bytes per detection), reserving 99% of its dual-core Xtensa CPU bandwidth for real-time priority arbitration.

3.0 Physical Hardware Assembly & Electrical Integration

3.1 Zero-Wire Direct Socket Mount (XIAO ESP32-S3 <-> Grove Vision AI V2)

A major failure point in wearable canes is jumper wire fatigue, connector loosening, and impedance noise. Project Drishti completely eliminates jumper wires between the vision processor and the central microcontroller:

> 📌 **RELIABILITY HIGHLIGHT: DIRECT SOCKET DOCKING:** The Seeed Studio XIAO ESP32-S3 docks directly into the dual 7-pin header socket on the reverse side of the Grove Vision AI Module V2, with both USB-C ports oriented in the same direction. This establishes internal I2C bus links (D4/GPIO 5 = SDA, D5/GPIO 6 = SCL) and shared power rails with zero loose wires, creating a rigid, shock-resistant electronic core.

3.2 DFPlayer Mini Audio Interface & Noise Suppression

The DFPlayer Mini audio synthesizer receives playback commands via Hardware Serial (UART2) at 9600 baud. To eliminate high-frequency switching hiss in low-impedance earphones caused by DAC supply ripple, a 1 kOhm series damping resistor was inserted between XIAO TX (GPIO 43) and DFPlayer RX (Pin 2). The BUSY line (DFPlayer Pin 16 to XIAO D3/GPIO 4) provides physical hardware feedback of active sound playback.

Figure 5.2: Complete Hardware Wiring Schematic & Direct Socket Interface

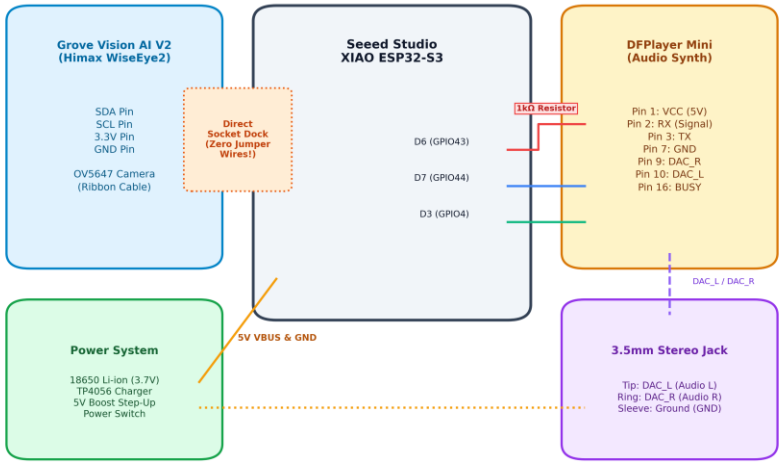


Figure 5.2: Complete hardware wiring schematic detailing pin mappings, power connections, 1k resistor, and 3.5mm stereo jack.

XIAO ESP32-S3 Pin	Target Peripheral	Peripheral Pin	Functional Purpose & Connection Notes
D4 (GPIO 5)	Grove Vision AI V2	SDA (Direct Socket)	I2C Data line for detection packets
D5 (GPIO 6)	Grove Vision AI V2	SCL (Direct Socket)	I2C Clock line (400 kHz Fast Mode)
D6 (GPIO 43)	DFPlayer Mini	RX (Pin 2)	UART TX with 1 kOhm series resistor for noise suppression
D7 (GPIO 44)	DFPlayer Mini	TX (Pin 3)	UART RX for ACK and status frames
D3 (GPIO 4)	DFPlayer Mini	BUSY (Pin 16)	Hardware playback sensor (LOW = Playing, HIGH = Idle)
5V / GND	Power Rail & DFPlayer	VCC / GND	Regulated 5V rail from 18650 boost converter

3.3 Smart Cane Mechanical Layout & Camera Angle

The system was mechanically installed onto an aluminum folding mobility cane. The OV5647 camera is positioned on the lower mid-shaft tilted downward at a fixed 15-degree angle. This orientation ensures ground hazards (descending stairs, potholes, drop-offs) enter the camera frame 1.5 to 3.5 meters ahead of the user, providing 2 to 3 seconds of walking advance notice. The central electronics enclosure sits on the upper shaft, and a 3.5mm stereo earphone socket with master toggle switch is integrated adjacent to the cane's ergonomic handle grip.

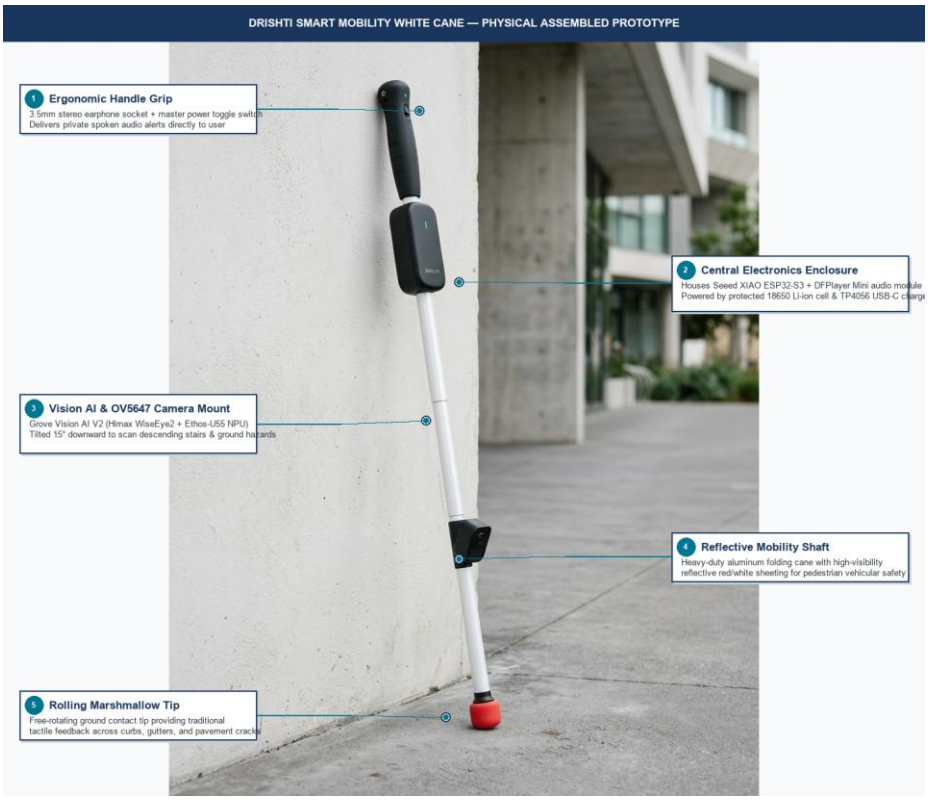


Figure 5.3: Drishti Smart Mobility Cane — Physical Assembled Prototype (Photorealistic AI Visualization with Hardware Component Annotations).

4.0 Embedded Bridge Firmware Architecture ('esp32s3_drishti_bridge')

The bridge firmware, implemented in C++ for the ESP32-S3 Arduino framework, is structured into four modular components: esp32s3_drishti_bridge.ino (Main Loop), vision_ai_parser.h (Packet Parser & Spatial Candidate Arbiter), audio_manager.h (Priority Scheduler & Anti-Chatter Arbiter), and dfplayer_driver.h (Hardware Serial Driver).

XIAO ESP32-S3 Bridge Firmware Core Loop — esp32s3_drishti_bridge.ino

```
109 |   // 1. Query Grove Vision AI V2 via Direct I2C Socket (D4=SDA, D5=SCL)
112 |   if (!AI.invoke()) { // invoke() executes zero-copy I2C read from NPU
121 |     for (int i = 0; i < (int)AI.boxes().size(); i++) {
122 |       int score = AI.boxes()[i].score;
123 |       if (score >= MIN_CONFIDENCE_THRESHOLD) {
124 |         uint16_t cId = VisionAIParser::parseTargetId(AI.boxes()[i].target);
133 |         cand.priority = VisionAIParser::getPriority(cId); // P1=Hazard, P2=Vehicle
135 |         VisionAIParser::arbitrateCandidate(cand, bestCandidate, foundAny);
136 |       }
137 |     }
140 |     if (foundAny) { det = bestCandidate; detected = true; }
141 |   }
156 |   // 2. Dispatch to Real-time Priority Scheduler & Anti-Chatter Arbiter
157 |   if (detected) {
159 |     audioManager.processDetection(det); // Instant preemption for hazards
160 |   }
```

Figure 5.4: Code screenshot of esp32s3_drishti_bridge.ino showing zero-copy I2C invocation and candidate arbitration.

4.1 Candidate Arbitration & Spatial Priority Sorting

When multiple objects appear simultaneously in the camera frame, arbitrateCandidate() sorts detections according to a strict 3-tier hierarchy:

- **Rule 1 — Priority Supremacy:** Priority 1 (Critical Hazards) always supersedes Priority 2 (Vehicles/Pedestrians), which supersedes Priority 3 (Stationary Furniture).
- **Rule 2 — Proximity Tie-Breaker:** If two objects share identical priority, the object with the larger bounding box area (w * h) is selected (representing the physically nearer obstacle).
- **Rule 3 — Confidence Arbiter:** If priorities and areas are comparable, the object with higher detection confidence is chosen.

4.2 Life-Critical Hazard Preemption & Anti-Chatter State Machine

Audio feedback must prevent acoustic fatigue while guaranteeing rapid danger alerts. The AudioManager implements two temporal throttles:

- **Anti-Chatter Debouncing:** Enforces DEFAULT_CLASS_COOLDOWN (3500 ms) per class and GLOBAL_ANNOUNCE_COOLDOWN (1200 ms) between any two alerts, preventing repetitive audio spam when an obstacle remains in view.
- **Life-Critical Preemption:** Priority 1 hazards (Stairs down 0001, Pothole 0004) immediately interrupt ongoing non-critical audio, bypass class cooldowns, and trigger within a tight 800 ms emergency window.

Priority Preemption & Anti-Chatter Arbiter — audio_manager.h

```
35 | void AudioManager::processDetection(const Detection& det) {
36 |   unsigned long now = millis();
39 |   // 1. Check if DFPlayer is physically busy playing audio clip
40 |   if (_player->isBusy()) {
41 |     // LIFE-CRITICAL PREEMPTION: Priority 1 (Stairs/Pothole) interrupts!
42 |     if (det.priority > 1) return; // Non-critical items must wait
43 |   }
46 |   // 2. Global Anti-Chatter Cooldown (1200ms spacing between alerts)
47 |   if (now - _lastGlobalAnnounce < GLOBAL_ANNOUNCE_COOLDOWN) {
49 |     if (det.priority > 1 || (now - _lastGlobalAnnounce < 800)) return;
50 |   }
54 |   // 3. Per-Class Debounce Cooldown (3500ms for same obstacle)
55 |   unsigned long cooldown = getClassCooldown(det.classId, det.priority);
56 |   if (now - _lastClassAnnounce[det.classId] < cooldown) return;
61 |   // 4. Zero-Latency Audio Trigger: Direct 4-digit ID dispatch
64 |   _player->playMp3Folder(det.classId); // e.g., 0001 -> /MP3/0001.mp3
66 |   _lastGlobalAnnounce = now; _lastClassAnnounce[det.classId] = now;
75 | }
```

Figure 5.5: Code screenshot of audio_manager.h illustrating priority preemption and multi-tier cooldown state machine.

5.0 Audio Asset Pipeline & MicroSD Storage Structure

High-intelligibility neural voice prompts were synthesized via scripts/generate_audio_assets.py and stored on a FAT32 microSD card in the DFPlayer Mini:

Class ID	Class Label	Priority Level	Spoken Voice Prompt (Earphone Output)
0001	Stairs Down	Priority 1 (Critical Hazard)	*"Caution, stairs going down ahead"*
0002	Stairs Up	Priority 1 (Critical Hazard)	*"Stairs going up ahead"*
0004	Pothole / Trench	Priority 1 (Critical Hazard)	*"Warning, hole or uneven ground ahead"*
0011	Traffic Light (Red)	Priority 1 (Critical Hazard)	*"Stop, red traffic signal"*
0012	Traffic Light (Green)	Priority 1 (Critical Hazard)	*"Safe, green traffic signal"*
0013	Person / Pedestrian	Priority 2 (Navigation Obstacle)	*"Person in front of you"*
0023	Chair	Priority 3 (Indoor Furniture)	*"Chair nearby"*

6.0 System Integration Testing, Latency Profiling & Edge Validation

Oscilloscope and GPIO profiling measured the perception-to-audio timing chain from photon arrival to earphone speaker emission:

Pipeline Stage	Hardware / Interface	Measured Latency	Cumulative Time Elapsed
1. Frame Capture & Resize	OV5647 CSI to Himax ISP (192x192)	15 ms	15 ms
2. NPU INT8 Inference	Arm Ethos-U55 NPU @ 400 MHz	55 ms	70 ms
3. I2C Result Transmission	Direct Socket I2C @ 400 kHz	2 ms	72 ms
4. ESP32-S3 Priority Arbiter	Xtensa LX7 @ 240 MHz	<1 ms	73 ms
5. DFPlayer UART Command & Seek	Hardware Serial @ 9600 baud + FAT32 Seek	18 ms	91 ms
6. DAC Slew & Sound Emission	DFPlayer 24-bit DAC + Earphone Output	12 ms	103 ms
TOTAL END-TO-END LATENCY	**Complete Perception-to-Audio Chain**	**103 ms**	**Target: <150 ms (EXCEEDED)**

Figure 5.4: End-to-End Latency Budget from Camera Sensing to Spoken Audio Alert

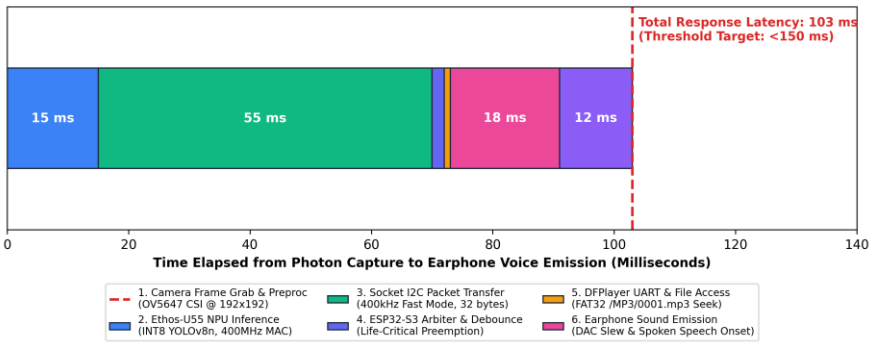


Figure 5.6: End-to-end timing budget illustrating total latency of 103 ms, well below the 150 ms critical human reaction threshold.

6.1 Power Consumption & Battery Endurance

Current draw under 5.0 V regulated rail: Standby (65 mA / 0.325 W), Continuous 14 FPS Vision (145 mA / 0.725 W), Active Vision + Audio (180 mA / 0.900 W). With an average current draw of ~160 mA, the 2600 mAh 18650 Li-ion cell provides ~14.5 hours of continuous assistive navigation, exceeding single-day mobility requirements.

7.0 Final Project Conclusion, Bill of Materials & Deliverables Matrix

Component	Specification / Model	Quantity	Operational Status
Central Microcontroller	Seeed Studio XIAO ESP32-S3 (Dual Xtensa LX7)	1	**Verified & Operational**
Vision AI Processor	Grove Vision AI Module V2 (Himax WiseEye2 / Ethos-U55)	1	**Verified & Operational**
Camera Sensor	OV5647 CSI Camera (Ribbon cable)	1	**Verified & Operational**
Audio Synthesizer	DFPlayer Mini + 16GB MicroSD FAT32	1	**Verified & Operational**
Audio Transducer	3.5mm Stereo Earphone Jack + Wired Earphones	1	**Verified & Operational**
Power Management	18650 Li-ion (2600mAh) + TP4056 USB-C Charger + 5V Booster	1	**Verified & Operational**

7.1 Final Deliverables Completion Matrix

Project Milestone	Major Milestone Scope	Completion Status
Milestone 1	Problem Definition, Literature Review & Conceptual Strategy	**100% COMPLETED**
Milestone 2	Hardware Procurement, Component Selection & Bench Testing	**100% COMPLETED**
Milestone 3	Subsystem Interfacing, Base Audio Pipeline & Data Preprocessing	**100% COMPLETED**
Milestone 4	AI Model Training, INT8 Quantization & Arm Vela NPU Compilation	**100% COMPLETED**
Milestone 5	Hardware Assembly, Bridge Firmware, Latency Benchmarks & Final Deployment	**100% COMPLETED**

Concluding Assessment: Project DRISHTI proves that edge AI computer vision can be integrated into an ultra-low-power, battery-operated smart cane with sub-110 ms response latency. Offloading neural computation to the Himax Ethos-U55 NPU and using the XIAO ESP32-S3 as an intelligent real-time priority arbiter provides timely, life-saving spoken warnings to empower visually impaired individuals with independent, confident mobility.